Leadership, Craftsmanship, and Execution



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The following is a memo that I sent to my team, the Engineering & Operations group at LinkedIn, as part of a series of weekly, Friday Dispatches. Given the nature of this particular memo, I thought that I would share more broadly to give folks a bit of perspective on how we think about the practice of engineering at LI.

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Folks,

LinkedIn's mission—to connect the world's professionals to make them more productive and successful—is deeply dependent upon our ability as a team of 1500+ engineers, in partnership with our peers across the company, to develop extremely complex, innovative technology at massive scale. You all prove to yourselves, to me, and to our members day-by-day that you are pretty damned good at this. But given your choice of work and the nature of the mental wiring that attracted you to the engineering profession in the first place, you probably walk around all day, all night identifying the broken and sub-optimal in nearly everything, knowing somewhere deep inside your soul that there must be a better way. And if you're like me, your case of *Engineer's Syndrome* consistently ignores the super awesome work that might have just flowed from your finger tips to production to our members, and has you focusing on the gazillion ways that things could have been done better.

Although I will occasionally try to remind you all that you're doing great work (***psst you are doing great work***), I don't want you to ever lose your desire to improve things via technology, including improvements to the way that we do our work here at LinkedIn. This is at the very core of what makes you an engineer. That said, I do want to give all of you a framework for harnessing your impulse to make things better and focusing on what is going to matter most to us as we continue to build one of the great, enduring technology companies of the 21st century.

Have a look at the Venn diagram above (or that's attached to this e-mail). There's a circle labeled **leadership**, one labeled **craftsmanship**, and another labeled **execution.** Engineering teams that produce the most long-term innovation and value are ones where these three characteristics of their output are balanced and in alignment.

Granted, it is possible to do good short-term work when one of these characteristics is lagging or missing. For example, you can execute well and get good short-term results while throwing craftsmanship out of the window, although that typically leads to very serious team and technology stability issues longer term. You can sometimes have good short-term execution and

craftsmanship in the absence of solid leadership just because the team is outstanding, although situations like this are almost never sustainable over the longer term. And if you have great leadership and craftsmanship, you might end up having a transformative personal experience building something really cool, but without execution, you are probably toast as soon as your stockpile of money runs out.

The holy grail is to have leadership, craftsmanship, and execution in balance, valued, and actively practiced throughout a technology organization at all levels.

Leadership as we typically define it is inspiring others to accomplish a common objective or mission. As an engineer you often inspire by setting a great example for your peers through the code you write or the technical decisions that you drive, by leverage that your work creates for others, by coaching and mentoring those with different experiences than yours, or by the impact that your work has on our members. As a manager your leadership responsibilities expand beyond owning a technical roadmap and deliverables to co-owing career development of the folks whom you manage, ensuring that everyone understands their career path, is recognized for the work they do, and has the appropriate resources available to them to do their job and to advance toward their long-term career objectives.

Craftsmanship is sometimes defined as the quality of design and workmanship shown in something made by hand. It is in some ways the most important aspect of our job as engineers, and yet one of the most difficult to precisely quantify. Fortunately, it is easy to spot and describe anti-patterns for engineering craftsmanship: unreadable, unmaintainable, untestable, poorly documented, non-reusable code; purposeless complexity and ugly systems architecture; crappy user experiences (which applies to APIs and infrastructure as easily as it does to consumer-facing apps); and brittle, flaky, unpredictable services that are difficult to troubleshoot and that take a long time to recover from an error state. There are some good resources out there on engineering craftsmanship, although perhaps the best at your disposal is your peers. There is an amazing group of folks here at LinkedIn with a wealth of backgrounds and experiences from which each of us can benefit to become better at our craft.

Execution is probably the thing that requires the least explanation. Simply put, it is getting shit done. That said, good execution is no easy thing. It requires solid collaboration, quantifying and measuring success, setting aggressive goals, and holding yourself and others accountable for meeting commitments. Moreover, good long-term execution requires a shared understanding of clearly articulated strategy (whether it is for a small project or a company-wide moon-shot), and a focus on balancing delivery of results for the short-term with creating highly-leveraged value for the long-term. Not at all easy, although something this team is better at than any other with which I've worked.

Valuing a balance of leadership, craftsmanship, and execution, and trying to pay attention to all three in the presence of your regularly scheduled daily circus is tough. I struggle constantly with getting the balance right. The struggle is worth having though, in fact a necessity, because it is literally the only way for us to fully deliver on LinkedIn's mission for our members, and the only way for us to be the type of engineering team that I want us to be. Our mission is too important, and our potential is too great to do anything less.

Have a great weekend!

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Ravi Shankar Rajan · 3rd+ 12y
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Agree with the three circles. But I believe the leadership circle encompasses the craftsmanship and execution circles. Leadership can be everywhere and it is required in everything right from crafting to executing. One more point that is important is that leaders need not always lead people; every single self organizing engineer can be a ...more

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Great post, Kevin. Thanks for sharing it publicly.

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James Gourgoutis · 3rd+ 11y
Development CNC Machinist at Astra

Leadership, craftsmanship, and quality are important in any endeavor, not just Engineering. It applies to design and product development, as well as all other roles, equally. I want us to be having this conversation across the whole company, not just within Engineering.

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Paul Murphy · 3rd+ 10y
Vice President, Major Accounts & Vertical Markets at Canon Solutions ...

Kevin,
I know this is a dated article, however this applies to creating new opportunities in legacy organizations. I have applied this principle and we have created a complete new approach to several very important markets. Thanks for the inspiration. Paul Murphy

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Three great elements very well articulated and amalgamated to arrive at the best final mission..Our mission is too important, and our potential is too great to do anything less.

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I like the "craftsmanship" category, and don't often hear about the importance of it. I think that the criticality of "elegance" in network engineering and operations is often overlooked, and it's hard in much the same way that "just because you can doesn't mean you should" is often a multi-faceted/multi-discipline challenge.

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Excellent post! Just love it!

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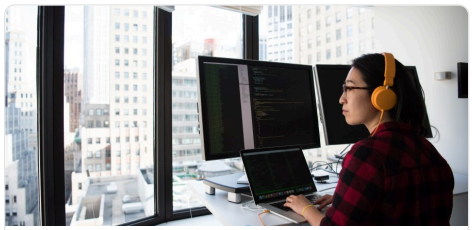
Kevin Scott

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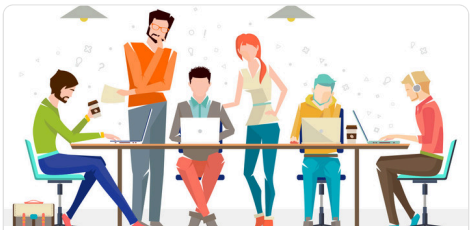
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